

Alexandru-Octavian Oprea

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Education

University of Freiburg

MSc, Earth and Planetary Science

Freiburg, Germany

Graduation Date: 09.2027

Relevant Coursework: Intro to Planetary Sciences, Geophysics, Petrophysics, Space Missions and Instrumentation, Scientific Computing, Data Analysis, Earth and Planetary Interiors, Impact Cratering and Shockwaves.

Jacobs University (Constructor University)

BSc, Earth and Environmental Sciences

Bremen, Germany

Graduation Date: 12.2024

Relevant Coursework: Intro to GIS, Geophysical Remote Sensing, Programming with Python, Marine Geophysics, Data Science for Earth Sciences, Modeling of Earth System Data.

National Computer College Tudor Vianu

Secondary Education: Mathematics and Computer Science

Bucharest, Romania

Graduation Date: 07.2021

Activities: FIRST Tech Challenge robotics; SolidWorks mechanical design and Java autonomous programming

Experience

University of Freiburg

Research Assistant (HiWi)

Freiburg, Germany

February 2026 – June 2026

- Built an equation-of-state database with Prof. Clemens Prescher using FastAPI and PostgreSQL, deployed with Docker.

Jacobs University

Guest Researcher

Bremen, Germany

May 2025 – August 2025

- Quantified trace Cr(VI) in seawater samples using UV digestion and standard addition voltammetry.
- Prepared calibration standards and buffers, and maintained a Chelex resin purification setup.

Jacobs University

Teaching Assistant

Bremen, Germany

September 2023 – February 2025

- Ran weekly Python tutorials on scientific computing, data analysis, and geospatial processing.
- Supported courses in remote sensing, marine geophysics, and Earth system modeling.
- Taught sonar data processing ahead of North Sea fieldwork; made tutorials on Sentinel-1 volcano monitoring.

Internships

German Aerospace Center (DLR), Institute of Planetary Research

Research Intern

Berlin, Germany

August 2026 – November 2026

- Running GAIA convection simulations of Ganymede's ice shell to study subsurface layering and resurfacing, and reproducing published crater-based layer thickness estimates.

Helmholtz-Zentrum Hereon

Physical Oceanography Intern

Geesthacht, Germany

June 2023 – August 2023

- Wrote Python scripts to detect and characterize marine heatwave events in climate datasets.

Skills & Interests

Technical: Python (NumPy, SciPy, pandas, matplotlib, xarray), GAIA, FastAPI, PostgreSQL, Docker, Git, QGIS, LaTeX, Linux

Languages: Romanian (Native), English (C2), German (B1)

Laboratory: Voltammetry, UV digestion, side scan sonar, satellite image processing

Interests: Planetary Geology, Icy Moons, Astrobiology, Remote Sensing, Astronomy